

TABLE II
Overview of the Features of Hardware Platforms for WMSNs

Device name	Manufacturer	Processor	Memory	Multimedia support	Wireless
Stargate	Crossbow	Intel PXA-255 Xscale processor at 400MHz	32MB flash, 64MB RAM	High computation power, embedded Linux OS	802.11 compact flash, 802.15.4 through MICA2/z interface
Imote2	Intel	32-bit PXA271 Marvell processor at 13MHz-416MHz	32MB flash, 64MB RAM	MMX co-processor for audio/video imaging acceleration	Integrated 802.15.4
CMUcam3	CMU	32-bit NXP LPC2106 microcontroller at 60MHz	128MB flash, 64MB RAM	On-board cc3 open source image-processing library	—
MeshEye	Stanford University	32-bit ARM7TDMI RISC processor at 55MHz	128MB flash, 64MB RAM	Multiple resolution support	—
WiCa	XNP and Philips Research	IC3D Xetal II processor at 84MHz	10MB RAM	Dedicated parallel processor, multiple camera modules	—
Cyclops	Agilent	8-bit ATMEL ATmega 128L microcontroller	512KB flash, 512KB RAM	On-board image processing, low power, cost and size	—

(Courtesy: Akyildiz et al.: Wireless multimedia sensor networks: applications and test beds)

A coordination sub-system coordinates the operation of different network devices by performing various tasks, such as network synchronisation and location management. A mobility actuation unit can enable the movement or manipulation of objects, like motor, to track the object. The whole system is powered by a compact power unit that may be supported by an energy-scavenging unit, such as solar cells. Table II gives an overview of the features of hardware platforms for WMSNs.

WMSN architecture

Fig. 3 shows the reference architecture of a WMSN, which is classified as single-tier and multi-tier architecture. Before we discuss further, it is important to know about the functionality of all components used in this architecture.

Standard audio and video sensors capture sound, still or moving images and videos of low resolution. Scalar sensors are another type of sensors that sense scalar data and physical attributes, such as temperature, humidity and pressure, and report measured values to their cluster head. These are typically resource-limited devices in terms of energy consumption, memory storage and processing capabilities.

Multimedia processing hubs behave as cluster heads. These devices have comparatively large computational resources and are suitable for aggregating multimedia streams from individual sensor nodes. This can be done by

various algorithms implemented in it. Algorithms are capable of managing flow of control (frames per second) by adding and discarding frames. Finally, it is capable of reducing both the dimensionality and volume of data conveyed to the sink and storage devices.

The storage hub allows data mining and feature-extraction algorithms to recognise essential characteristics of the event, even before the data is sent to the end user. The sink is a common data-gathering point of IP-less networks. It also supplies filtered data to the user at the remote end, gathered by a WMSN. Multiple sinks may be required in a large or heterogeneous network.

The gateway creates a bridge between an IP-less network and IP based network. It is an IP-addressable component of the WMSN. It gives permission to the user to monitor and control the WMSN from a remote location as it has an IP address and the capability to create a connection with the Internet.

At the end, users are at the highest end of the hierarchy. They are typical-

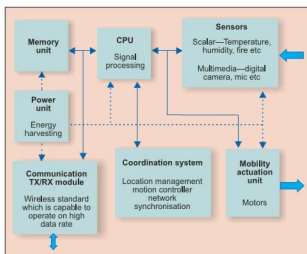


Fig. 2: Block diagram of a multimedia sensor node

ly identified through their IP addresses and run application-level protocols and software that assign queries and display results obtained from the WMSN.

Single-tier means the network is without hierarchy, and cluster is directly connected with the sink. Figs 3(a) and 3(b) show a single-tier architecture, which can be arranged in a flat or clustered manner. Flat topology means all nodes have essentially the same role.

In cluster topology, a group of nodes form a cluster, and the geographical target area is divided into numbers of clusters. Each cluster has one cluster head and one cluster member. In a cluster, two members cannot communicate directly, as shown in Fig. 3(b).